SQL Assignment Questions (Based on SELECT, FROM, WHERE, *, IN, NOT IN, AND, OR, %, >, <, =, MOD)

Instructions: - These questions are based only on the topics mentioned above. - No wildcards or pattern matching are involved. - Focus on logical thinking — questions involving MOD, comparisons, and conditional reasoning are included. - Each question is written in plain English (not in SQL syntax form). - Difficulty levels: Easy / Medium / Hard / Interview.

Easy Level

1. Display all information about every product in the database.
2. Find all vendors who have a specific vendor ID equal to 5.
3. Show the list of items from vendor inventory that have a quantity greater than 10.
4. Retrieve details of all purchases made for product IDs 2 or 3.
5. Identify all products that do not belong to category 1, 2, or 3.
6. Display all booths that have a price level of 'A'.
7. Show booth assignments for the market date of '2019-04-03'.
8. Find the customers who live in the ZIP code 22821.
9. Identify vendor inventory entries with a price below 5.
10. Find all products with quantity between 5 and 15.

Medium Level

1. List all vendors whose ID is either 1, 3, or 7.
2. Show all customer purchases where the customer ID is not 10 or 12.
3. Retrieve inventory records from vendor 7 on the market date '2019-07-03'.
4. Identify all booths that are of type 'Standard' or 'Small'.
5. Find inventory records where the quantity is less than 8 or the price is greater than 8.
6. Retrieve customers from ZIP codes 22801 or 22821.
7. Find all vendor inventories where quantity is greater than 10 or the price is less than 4.
8. Find purchases where the product ID is among 1, 4, 7 and vendor ID is 3.
9. Identify booth numbers greater than 5 but less than 10.
10. Find vendors whose IDs are even numbers.
11. Display all vendors with odd-numbered vendor IDs.
12. Identify products that are in category 2 and have size 'large'.
13. List customers whose ID is greater than 5 but less than 15.
14. Find all inventory entries where the quantity is a multiple of 5.
15. Retrieve purchases made on '2019-07-01' for products 2, 5, or 9.
16. Show all booths that have a type 'Standard' and a price level 'B'.
17. List vendors that are not 2, 4, or 6 and belong to 'Eggs & Meats'.
18. Identify all products with an ID below 10 or category above 5.
19. Retrieve vendor inventories where the price is 6.99 and quantity is odd.
20. Find all vendors whose IDs are not in 3 or 5 but less than 8.

Hard Level

1. Find all inventory rows where the quantity is greater than 0, less than 10, and the product ID is among 1, 2, 3, 4, or 5.
2. Identify all product IDs that are divisible by 3.
3. Determine all product IDs that are divisible by both 2 and 3.
4. Find vendor booths for a given date where booth numbers are not among 1, 2, or 3 and less than 10.
5. Retrieve all vendor inventory entries where price is between 5 and 7, and quantity is not zero.
6. Find all vendor inventories where the quantity is a multiple of 4 or the price is above 8.
7. Identify products that belong to category 1 or have an ID greater than 20.
8. Determine all vendors where the vendor ID divided by 5 leaves no remainder.
9. Find customers whose IDs are odd numbers.
10. Retrieve all records from vendor inventory where quantity is even and price is above 5.
11. Identify all vendors whose IDs are divisible by 3 but not by 2.
12. Find all vendors whose IDs leave a remainder of 1 when divided by 3.
13. Identify booth numbers that are multiples of 3 and belong to vendor 2.
14. Retrieve all inventory entries where quantity is greater than 12 and less than 20, and the price equals 6.99.
15. Determine all customer purchases where quantity is greater than 2 and product ID is either 1 or 3.
16. Find all vendors whose ID is less than 5 and vendor type is 'Fresh Variety: Veggies & More'.
17. Identify all booths that have numbers between 2 and 8 and belong to vendors 1 or 3.
18. Find all vendor IDs that are divisible by 4 but not by 8.
19. Determine products whose ID when divided by 5 gives a remainder of 2.
20. Find all vendors whose ID is either a multiple of 5 or less than 10 but not in the set (2,7).

End of SQL Assignment Question Set

Solutions (SQL)

1. SELECT * FROM product;

2. SELECT * FROM vendor WHERE vendor_id = 5;

3. SELECT * FROM vendor_inventory WHERE quantity > 10;

4. SELECT * FROM customer_purchases WHERE product_id IN (2, 3);

5. SELECT * FROM product WHERE product_category_id NOT IN (1, 2, 3);

6. SELECT * FROM booth WHERE booth_price_level = 'A';

7. SELECT * FROM vendor_booth_assignments WHERE market_date = '2019-04-03';

8. SELECT * FROM customer WHERE customer_zip = 22821;

9. SELECT * FROM vendor_inventory WHERE original_price < 5.00;
10. SELECT product_id, quantity FROM vendor_inventory WHERE quantity > 5 AND quantity < 15;
11. SELECT * FROM vendor WHERE vendor_id IN (1, 3, 7);
12. SELECT * FROM customer_purchases WHERE customer_id NOT IN (10, 12);
13. SELECT * FROM vendor_inventory WHERE market_date = '2019-07-03' AND vendor_id = 7;
14. SELECT * FROM booth WHERE booth_type IN ('Standard', 'Small');
15. SELECT * FROM vendor_inventory WHERE quantity < 8 OR original_price > 8.00;
16. SELECT * FROM customer WHERE customer_zip IN (22801, 22821);
17. SELECT * FROM vendor_inventory WHERE quantity > 10 OR original_price < 4.00;
18. SELECT * FROM customer_purchases WHERE product_id IN (1, 4, 7) AND vendor_id = 3;
19. SELECT * FROM vendor_booth_assignments WHERE booth_number > 5 AND booth_number < 10;
20. SELECT * FROM vendor WHERE MOD(vendor_id, 2) = 0;
21. SELECT * FROM vendor WHERE MOD(vendor_id, 2) = 1;
22. SELECT * FROM product WHERE product_category_id = 2 AND product_size = 'large';
23. SELECT * FROM customer WHERE customer_id > 5 AND customer_id < 15;
24. SELECT * FROM vendor_inventory WHERE MOD(quantity, 5) = 0;
25. SELECT * FROM customer_purchases WHERE market_date = '2019-07-01' AND product_id IN (2, 5, 9);
26. SELECT * FROM booth WHERE booth_type = 'Standard' AND booth_price_level = 'B';
27. SELECT * FROM vendor WHERE vendor_id NOT IN (2, 4, 6) AND vendor_type = 'Eggs & Meats';
28. SELECT * FROM product WHERE product_id < 10 OR product_category_id > 5;
29. SELECT * FROM vendor_inventory WHERE original_price = 6.99 AND MOD(quantity, 2) = 1;
30. SELECT * FROM vendor WHERE vendor_id NOT IN (3, 5) AND vendor_id < 8;

31. SELECT * FROM vendor_inventory WHERE quantity > 0 AND quantity < 10 AND product_id IN (1,2,3,4,5);

32. SELECT product_id FROM product WHERE MOD(product_id, 3) = 0;

33. SELECT product_id FROM product WHERE MOD(product_id, 2) = 0 AND MOD(product_id, 3) = 0;

34. SELECT * FROM vendor_booth_assignments WHERE market_date = '2019-04-03' AND booth_number NOT IN (1,2,3) AND booth_number < 10;

35. SELECT * FROM vendor_inventory WHERE original_price >= 5 AND original_price <= 7 AND quantity NOT IN (0);

36. SELECT * FROM vendor_inventory WHERE MOD(quantity, 4) = 0 OR original_price > 8.00;

37. SELECT * FROM product WHERE product_category_id = 1 OR product_id > 20;

38. SELECT * FROM vendor WHERE MOD(vendor_id, 5) = 0;

39. SELECT * FROM customer WHERE MOD(customer_id, 2) = 1;

40. SELECT * FROM vendor_inventory WHERE MOD(quantity, 2) = 0 AND original_price > 5.00;

41. SELECT * FROM vendor_inventory WHERE quantity > 0 AND quantity < 10 AND product_id IN (1,2,3,4,5);

42. SELECT * FROM product WHERE product_category_id = 1 OR product_id > 20;

43. SELECT * FROM vendor_booth_assignments WHERE market_date = '2019-04-03' AND booth_number NOT IN (1,2,3) AND booth_number < 10;

44. SELECT * FROM vendor_inventory WHERE original_price >= 5 AND original_price <= 7 AND quantity NOT IN (0);

45. SELECT product_id FROM product WHERE MOD(product_id, 2) = 0 AND MOD(product_id, 3) = 0;

46. SELECT * FROM customer_purchases WHERE customer_id IN (2,4,6) OR (product_id IN (1,2) AND quantity > 3);

47. SELECT * FROM booth WHERE booth_price_level = 'B' OR booth_type = 'Small';

48. SELECT * FROM market_date_info WHERE market_rain_flag = 1 OR (market_snow_flag = 1 AND market_week > 10);

49. SELECT * FROM vendor_inventory WHERE MOD(quantity, 4) = 0 OR original_price > 8.00;

50. SELECT * FROM vendor WHERE MOD(vendor_id, 5) = 0 OR (vendor_id < 10 AND vendor_id NOT IN (2,7));